

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 1.2 INDICES: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Final Explanation: Why do scientists, economists and farmers use small superscript numbers? Look back at the two cards from Lesson 1: Card A shows Kenya's 2019 Census figure 47,564,296 written in full, and Card B shows 4.8×10^7 . Answer every section in the space under the prompt. Write in full sentences, show your working line by line, and name the law of indices you use at each step. Use the ideas from all eight lessons: repeated multiplication, the product law, the quotient law, the zero-index law and the power-of-a-power law. Where a question asks you to explain, give a reason, not just an answer.

Part 1: Reading the small superscript number

Look again at Card B: 4.8×10^7 . (a) Name the base and the index (power) in 10^7 , and say what each one tells you. (b) Write 10^7 as repeated multiplication and give its value in ordinary form. (c) Explain why Card B uses a superscript 7 when Card A is the Census figure 47,564,296. Why is Card B shorter, easier to compare and less likely to be misread? (d) Write 2^5 as repeated multiplication and evaluate it.

Write your answer here:

Part 2: The product law and projecting the population

(a) Expand $10^2 \times 10^3$ fully as repeated multiplication, count all the factors, and write the answer as a single power of 10. (b) Use your working to explain in words why multiplying powers with the same base means adding the indices. Write the general law using a , m and n . (c) Suppose Kenya's population of about 4.8×10^7 were projected to grow by a factor of 10^2 . Use the product law to write the projected population in the form $a \times 10^n$. Explain why this is quicker than multiplying out the full numbers.

Write your answer here:

Part 3: Division and the surprising power zero

(a) Expand $2^5 \div 2^2$ as a fraction, cancel the common factors, and write the answer as a single power. State the quotient law for $a^m \div a^n$. **(b)** Now work out $2^3 \div 2^3$ in two ways: first by the quotient law, then by ordinary arithmetic. What do the two results tell you about 2^0 ? **(c)** A classmate says: 'Zero copies of a number must be zero, so $a^0 = 0$.' Use your results from (b) to explain whether your classmate is right.

Write your answer here:

Part 4: Power of a power and multi-step problems

(a) Expand $(2^3)^2$ fully, count the factors of 2, and write the result as a single power. State the law for $(a^m)^n$. (b) A smartphone stores about 1 GB $\approx 2^{30}$ bytes, and 1 TB = $(2^{10})^3$. Use your law to write 1 TB as a single power of 2. (c) Simplify $(3^2 \times 3^5) \div 3^4$ and evaluate your answer. List the law you use at each step. (d) Simplify $((2^3)^2 \times 2^4) \div 2^6$ and evaluate your answer. Explain why you must deal with the bracket first.

Write your answer here:

Part 5: Answering the Driving Question

Driving Question: Why do scientists, economists and farmers use small superscript numbers to write huge or tiny quantities, and what rules make those compact symbols so powerful for computation? Write a paragraph of 6 to 9 sentences to explain Card B (4.8×10^7) to a younger learner who has never seen index notation. Your paragraph must (i) explain what the 7 means, (ii) name all four laws of indices and say what each does to the indices, (iii) explain the single big idea that all four laws come from, and (iv) give one real-life Kenyan application of index laws, for example from census data, the national budget, mobile data storage or M-Pesa.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Meaning of index notation	Correctly names base and index, expands powers as repeated multiplication with correct values, and clearly explains why Card B is more compact and less error-prone than Card A.	Names base and index and expands correctly, with only a brief or partial explanation of why index form is compact.	Confuses base and index, or expands incorrectly, or gives no explanation of the compact form.
Applying the four laws accurately	Applies the product, quotient, zero-index and power-of-a-power laws correctly in every part, including the multi-step problems, with the right order of steps and correct final values.	Applies most laws correctly with minor slips in the multi-step problems or in evaluating.	Applies laws incorrectly, for example multiplying indices in the product law, or cannot combine laws in a sequence.
Reasoning from repeated multiplication	Justifies each law by expanding and counting or cancelling factors, and explains why $a^0 = 1$ using both the quotient law and arithmetic.	Justifies some laws by expansion but states others without a reason.	States the laws as rules to memorise with no justification, or believes $a^0 = 0$.
Communication and Kenyan context	Writes clearly, shows each line of working, names the law used at each step, and links the ideas accurately to the Census figure and at least one other	Working is mostly clear and the laws are named, with a general or partly developed Kenyan link.	Working is missing or hard to follow, laws are not named, and there is no relevant Kenyan link.

	Kenyan application.		
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ATTRIBUTION AND LICENSING

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